

Inference for Categorical Time Series with Missing Observations

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27.04.2026

Motivation

- ▶ Many real-world time series are **categorical**:
 - ▶ Credit ratings (AAA, AA, ...)
 - ▶ Weather states (sunny, rainy, ...)
 - ▶ Survey responses (agree, neutral, disagree)
- ▶ Classical time series methods assume **numeric structure**
- ▶ $\Rightarrow$ Need new tools for:
 - ▶ Computing proper tests for marginal features
 - ▶ Measuring temporal dependence without bias

Categorical Time Series

Definition

A categorical time series (X_t) takes values in a finite set

$$S = \{s_0, \dots, s_m\}.$$

Two types of categorical data:

- ▶ **Nominal:** no natural ordering
- ▶ **Ordinal:** inherent ordering

Key implication:

- ▶ Available statistical tools depend on structure of S

Marginal Distributions

Central object: marginal probabilities

Nominal data:

$$p_i = P(X_t = s_i)$$

- ▶ Probability Mass Function (PMF)

Ordinal data:

$$f_i = P(X_t \leq s_i)$$

- ▶ Cumulative Distribution Function (CDF)

Key difference:

- ▶ Ordinal $\Rightarrow$ uses ordering
- ▶ Nominal $\Rightarrow$ no cumulative structure

Location Measures

Nominal data:

- ▶ No ordering $\Rightarrow$ no mean/median
- ▶ Use **mode**:

$$\text{Mod}(X_t) = \arg \max_{s_i} P(X_t = s_i)$$

Ordinal data:

- ▶ Ordering allows additional measures
- ▶ **Median**:

$$\text{Med}(X_t) = \min\{s_i : F(s_i) \geq 1/2\}$$

Interpretation:

- ▶ Mode = most frequent category
- ▶ Median = central position in ordering

Dispersion Measures

Different notions of variability $\Rightarrow$ Same for minimal, different for maximal variability

Ordinal: Index of Ordinal Variation (IOV)

$$\text{IOV}(\mathbf{f}) = \frac{4}{m} \sum_{i=0}^{m-1} f_i(1 - f_i)$$

Nominal: Index of Qualitative Variation (IQV)

$$\text{IQV}(\mathbf{p}) = \frac{m+1}{m} \left(1 - \sum_{i=0}^m p_i^2 \right)$$

- Both scaled to $[0, 1]$

Stationarity

Why do we need stationarity?

- ▶ Stable probabilistic structure over time
- ▶ Enables estimation and asymptotics

Strict Stationarity

$$F_{X_t, \dots, X_{t+j}} = F_{X_{t+h}, \dots, X_{t+h+j}}$$

for all t, j, h .

- ▶ Distribution invariant under time shifts

Temporal Dependence

Goal: capture dependence across time

Problem:

- ▶ No meaningful arithmetic operations
- ▶ Classical ACF not applicable

Solution:

- ▶ Work with joint probabilities

Bivariate Probabilities

$$p_{ij}(h) = P(X_t = s_i, X_{t+h} = s_j), \quad f_{ij}(h) = P(X_t \leq s_i, X_{t+h} \leq s_j)$$

- ▶ Independence $\Rightarrow p_{ij}(h) = p_i p_j$ and $f_{ij}(h) = f_i f_j$

Idea: measure deviation from independence

$$\kappa(h)_{\text{nom}} = \frac{\sum_{j=0}^m (p_{jj}(h) - p_j^2)}{1 - \sum_{i=0}^m p_i^2} \quad \kappa(h)_{\text{ord}} = \frac{\sum_{j=0}^m (f_{jj}(h) - f_j^2)}{1 - \sum_{i=0}^m f_i(1 - f_i)}$$

- ▶ $\kappa = 1$: perfect persistence
- ▶ $\kappa = 0$: independence
- ▶ $\kappa < 0$: tendency to switch states

Interpretation:

- ▶ Probability of staying in same category
- ▶ Relative to independence benchmark

Estimation of Marginal Probabilities

Idea: use of indicator functions

$$Y_{t,i} = \mathbb{I}\{X_t = s_i\} \quad \text{and} \quad Z_{t,i} = \mathbb{I}\{X_t \leq s_i\}$$

Then estimators are given by:

- ▶ $\hat{p}_i = \frac{1}{n} \sum_{t=1}^n Y_{t,i}$
- ▶ $\hat{f}_i = \frac{1}{n} \sum_{t=1}^n Z_{t,i}$

Intuition:

⇒ we just count relative frequencies

From here we mainly look at ordinal data, but the nominal case is in many ways analogous.

Mixing Assumptions

Idea

The Central Limit Theorem (CLT) provides asymptotic normality and enables statistical inference.

Problem

Classical CLT results assume independence, but in practice data may exhibit serial dependence.

Solution

To handle dependence, we impose **mixing conditions**:

- ▶ α -mixing quantifies the decay of dependence between past and future observations
- ▶ Exponential decay of the mixing coefficients allows a CLT for dependent data

CLT under serial dependence

Result

$$\sqrt{n}(\hat{f} - f) \Rightarrow \mathcal{N}(0, \mathbf{\Sigma})$$

where the elements of $\mathbf{\Sigma}$ are given by:

$$\sigma_{ij} = \underbrace{(f_{\min\{i,j\}} - f_i f_j)}_{\text{lag}(0)} + \sum_{h=1}^{\infty} \underbrace{(f_{ij}(h) + f_{ji}(h) - 2f_i f_j)}_{\text{lag}(h)}.$$

Note that $\text{lag}(h) \rightarrow 0$ as $h \rightarrow \infty$.

CLT's for marginal estimators

Idea

IOV is function of the marginal probabilities

⇒ we can apply the **delta method** to obtain their asymptotic distributions

Results

- Asymptotic variance

$$\sqrt{n}\text{Var}(I\hat{O}V) \stackrel{\text{asy}}{=} \frac{16}{m^2} \sum_{i,j=0}^{m-1} (1 - 2f_i)(1 - 2f_j)\sigma_{ij}$$

- Bias to order n^{-1} :

$$n\text{Bias}(I\hat{O}V) = -\frac{4}{m} \sum_{i=0}^{m-1} \sigma_{ii}$$

Missingness - Relevance

Amplitude Modulation

Idea

Missingness can be modeled as a binary process (O_t) , that *modulates* the *amplitude* of the actual process:

$O_t = 1$ if X_t is observed and $O_t = 0$ if X_t is missing

The observed process is then given by $(O_t X_t)$.

Assumptions

- ▶ (O_t) is stationary with $P(O_t = 1) = \pi > 0$ and joint moments $E[O_t O_{t+h}] = \pi(h)$
- ▶ (O_t) is α -mixing with exponential decay

Missingness Mechanisms

Originally developed for cross-sectional data, but also relevant for time series:

- ▶ **MCAR:** Missing Completely At Random

$$P(O_t = 1 | X_t, X_{t-1}, \dots) = P(O_t = 1)$$

- ▶ **MAR:** Missing At Random

$$P(O_t = 1 | X_t, X_{t-1}, \dots) = P(O_t = 1 | X_{t-1}, \dots)$$

- ▶ **MNAR:** Missing Not at Random

$$P(O_t = 1 | X_t, X_{t-1}, \dots) \neq P(O_t = 1 | X_{t-1}, \dots)$$

From here on, unless stated otherwise, we assume MCAR.

Naive vs corrected estimators

Naive estimator

$$\hat{\mathbf{f}} = \frac{1}{n} \sum_{t=1}^n O_t \mathbf{Z}_t$$

⇒ biased and inconsistent due to missingness

Corrected estimator

$$\hat{\mathbf{f}}^* = \frac{1}{n} \frac{\sum_{t=1}^n O_t \mathbf{Z}_t}{\sum_{t=1}^n O_t}$$

Asymptotic properties of the corrected estimator

Result

$$\sqrt{n}(\hat{\mathbf{f}}^* - \mathbf{f}) \Rightarrow \mathcal{N}(0, \mathbf{\Sigma}^*)$$

where the elements of $\mathbf{\Sigma}^*$ are given by:

$$\sigma_{ij}^* = \frac{1}{\pi}(f_{\min\{i,j\}} - f_i f_j) + \frac{1}{\pi^2} \sum_{h=1}^{\infty} \pi(h)(f_{ij}(h) + f_{ji}(h) - 2f_i f_j)$$